

Software X: apsimNGpy Example

Magala et al. (2025)

1 Listing 1

```
from apsimNGpy.core.apsim import ApsimModel
from pathlib import Path

# Instantiate the model
model = ApsimModel(model='Maize', out_path='new_maize.apsimx')
# Edit the planting density in the manager node
model.edit_model(model_type='Models.Manager', model_name='Sow using a variable rule', Population=12)
# Download and replace weather data automatically
lonlat = (-93.44, 41.1234)
model.get_weather_from_web(lonlat=lonlat, start=1981, end=2022, source='daymet')
# Change the simulation start and end date
model.edit_model(model_type='Models.Clock', model_name='Clock',
                  start='1990-01-01', end='2021-12-31') # ISO 8601 format (YYYY-MM-DD)

# Run the model
model.run(report_name='Report')
# Get results
df = model.results
# Save the modified model
filename = str(Path(f"my_edited_maize_model_at_{lonlat}.apsimx").resolve())
model.save(file_name=filename)
```

2 Listing 2

```
from pathlib import Path
import os
from constants import custom_colors
from apsimNGpy.core.experimentmanager import ExperimentManager as Experiment

if __name__ == '__main__':
    wd = Path(__file__).parent / 'apsimx'
    wd.mkdir(exist_ok=True)
    os.chdir(wd)

    experiment = Experiment(model='Maize', out_path=(wd / 'out_maizer.apsimx').resolve())
    experiment.init_experiment()
    experiment.add_factor('[Fertilise at sowing].Script.Amount = 0, 300')
    experiment.add_factor('[Sow using a variable rule].Script.CultivarName = Dekalb_XL82, Melkassa, B_110, Pioneer_34K77')
    # Run the experiment
    experiment.run()
    df = experiment.results
    g = experiment.cat_plot(
        x='Amount', y='Yield', hue='CultivarName',
        kind='bar', height=8, aspect=1, palette=custom_colors,
    )
    # In Jupiter notebook, the graph is displayed even if show=False
    experiment.render_plot(
        save_as=(wd / 'experiment.png').resolve(),
        dpi=600, show=True, ylabel='Simulated corn grain yield (kg ha$^{-1}$)',
        xlabel='Nitrogen fertilizer (kg ha$^{-1}$)',
    )
```

3 Listing 3

```
from apsimNGpy.core.apsim import ApsimModel
from pathlib import Path
# Instantiate the model
model = ApsimModel(model='Maize', out_path='new_maize.apsimx')
```

